MILITARY SPECIFICATION

COMPOUNDING OIL

This amendment forms a part of Military Specification MIL-C-23947(Wep), dated 19 August 1964, and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment basic specification MIL-C-23947(Wep) is changed to read MIL-C-23947(AS).

Page 1, title: Delete", FOR SOLID PROPELLANT GRAINS MARK 75 AND MARK 76".

Page 1, paragraph 1.1: Line 2, delete "ail" and substitute "oil". Line 3, delete "the Mk 75 and Mk 76".

Page 1, paragraph 2.1, under "Standards":

Delete:	"MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes:
Substitute:	"MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective"
Add:	"Federal FED-STD-102	Preservation, Packaging, and Packing Levels".

Page 2, paragraph 3.1: Delete second and third sentences.

Page 2, paragraph 3.2, Table I, Viscosity Limitations: Change "153" to "150" and change "159" to "160.

Page 3, paragraph 4.3: Change last sentence to read "If manufacture is by batch process, batches may be combined to form a lot provided not more than one lot of each ingredient and no change in formulation or process is used." .

Page 3, paragraph 4.5: Delete and substitute:
"4.5 Quality Conformance Inspection. -The following procedures shall be performed to determine compliance with section 3. Other test methods may be used if they offer assurance of equal results. If test methods differ from those specified herein, a copy of those methods or reference to available sources for the methods shall be furnished to the procuring activity." .

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Page 3, paragraph 4.5.1: Delete and substitute:
"4.5.1 Sampling. -Sampling for quality conformance inspection shall be in accordance with Standard MIL-STD-414, Inspection Level IV.".

Page 4, paragraph 4.6(d): Change "nearest 0.25°F" to "nearest 0.2°F ".

Page 4, paragraph 4.7: Delete and substitute:
"4.7 SUS Viscosity at 100°F.

4.7.1 Preparation. -The following operations shall be performed in preparation for this determination:

(a) Rinse all non-water-soluble liquids from a size 200 viscometer tube using the proper solvent.

(b) Fill the viscometer tube with chromic acid and place in a 2-liter beaker.

(c) Add water to the beaker until at least 3/4 of the viscometer tube is covered.

(d) Heat this water to boiling and allow it to boil for at least 15 minutes.

(e) Remove the viscometer tube from beaker and rinse thoroughly with distilled water.

(f) Use a vacuum to pull the air through. Thoroughly air dry the viscometer tube.

4.7.2 Calibration. -The following calibration shall be performed in preparation for this determination:

(a) Using the invert and fill method, fill the viscometer tube with the standard oil.

(b) Place the viscometer tube in a 100 ± 1°F oil bath and allow 20 minutes minimum for temperatures to equalize.

(c) Using the suction bulb, raise the level of the oil above the top mark on the viscometer tube.

(d) Allow the oil level to fall and as the meniscus flows past the top mark on the viscometer tube, start the timer.

(e) As the meniscus flows past the bottom mark on the viscometer tube, stop the timer and record the time interval.

(f) Repeat this test until three time intervals agree within 1 second and the flow time of each run is a minimum of 200 seconds.

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(g) Calculate the viscometer coefficient (K) as follows:

$K = tv$

where: t = efflex time of standard oil in seconds.
v = viscosity of standard oil at temperature used.

The K calculated is for the particular viscometer used, and is accurate only for the temperature at which it was calibrated. Each viscometer has to be calibrated individually.

4.7.3 Conversion Factors. -For converting Kinematic viscosity to:

(a) centistokes $n_L = Kt_L$

(b) centipoises $n_p = Kt_p d_p$

(c) SUS SUS = centistokes x 4.635

where: t_L and t_p = efflux time in seconds

d_p = density

K = tube coefficent

4.7.4 Procedure

(a) Repeat steps contained in 4.7.2 using sample to be run.

(b) After the viscosity has been calculated for the Kinematic viscosity, this answer must be converted to SUS (Saybolt Universal Seconds) by using the conversion factors per 4.7.3.".

Page 6, paragraph 4.9: Add ", or equivalent.".

Page 7, paragraph 4.9.2: Delete and substitute the following:

"4.9.2 Procedure

(a) Place a drop or two of the sample on the lower prism. Close the prisms by means of the screw head.

(b) By means of screw on the left of the instrument move the alidade (the moving part) until the field of vision shows the boundary between light and shade just intersecting the crosshair.

(c) Turn the screw on the right of the insturment until the boundary between light and shade is as sharp as possible. Readjust the left screw if necessary.

(d) Read the refractive index directly from the sector using the lens. Repeat the adjustment two or three times and report the average of the values obtained."

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Page 8: Add,
"4.10 Preservation, Packaging, and Packing. -Preservation, packaging, and packing shall conform to section 5.".

Page 8, paragraph 5.1: Delete "or as specified (see 6.2)." and substitute ", as defined in Federal Standard FED-STD-102, or as specified (see 6.2).". Add "The requirements of section 5 apply only to direct purchases by or direct shipments to the Government.".

Page 8, paragraph 5.2: Delete "or as specified (see 6.2)," and substitute ", as defined in Federal Standard FED-STD-102, or as specified (see 6.2).".

Page 9, paragraph 6.1 : Following "motors", delete "Mk 38 and Mk 39".

Page 9, paragraph 6.2: Change "shall specify" to "may specify".

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